

Supplementary Materials for:

“Dual-Attention Deep Gaussian Processes for Multi-Objective Robust Parameter Design”

A Derivation of the Variational ELBO for Sparse GPs

This appendix provides a detailed derivation of the Evidence Lower Bound (ELBO) (Equation 4) described in Section 2.2. We consider the augmented GP model by introducing M inducing points $\mathbf{Z}$ and corresponding inducing variables $\mathbf{u}$. The joint probability of the observations $\mathbf{y}$, latent functions $\mathbf{f}$, and inducing variables $\mathbf{u}$ is given by:

$$p(\mathbf{y}, \mathbf{f}, \mathbf{u}) = p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u}), \quad (\text{A.1})$$

where $p(\mathbf{u}) = \mathcal{N}(\mathbf{0}, \mathbf{K}_{\mathbf{uu}})$ is the prior over inducing variables, and $p(\mathbf{f}|\mathbf{u})$ follows the standard GP conditional distribution. To approximate the intractable posterior $p(\mathbf{f}, \mathbf{u}|\mathbf{y})$, we propose a variational distribution $q(\mathbf{f}, \mathbf{u})$. Following the framework of Titsias (2009), we enforce the variational posterior to factorize as:

$$q(\mathbf{f}, \mathbf{u}) = p(\mathbf{f}|\mathbf{u})q(\mathbf{u}), \quad (\text{A.2})$$

where $q(\mathbf{u}) = \mathcal{N}(\mathbf{m}, \mathbf{S})$ is a free variational distribution to be optimized, and $p(\mathbf{f}|\mathbf{u})$ is fixed to the model prior conditionals. This corresponds to Equation (4) in the main text. **Accordingly, the goal** of variational inference is to minimize the Kullback-Leibler (KL) divergence between the approximate posterior and the true posterior, which is equivalent to maximizing the ELBO:

$$\mathcal{L} = \iint q(\mathbf{f}, \mathbf{u}) \log \frac{p(\mathbf{y}, \mathbf{f}, \mathbf{u})}{q(\mathbf{f}, \mathbf{u})} d\mathbf{f}d\mathbf{u} \quad (\text{A.3})$$

Substituting (A.1) and (A.2) into (A.3), the ELBO can be expanded as:

$$\begin{aligned} \mathcal{L} &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \log \frac{p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})}{p(\mathbf{f}|\mathbf{u})q(\mathbf{u})} d\mathbf{f}d\mathbf{u} \\ &= \iint p(\mathbf{f}|\mathbf{u})q(\mathbf{u}) \left[\log p(\mathbf{y}|\mathbf{f}) + \log \frac{p(\mathbf{u})}{q(\mathbf{u})} \right] d\mathbf{f}d\mathbf{u}. \end{aligned} \quad (\text{A.4})$$

The term $p(\mathbf{f}|\mathbf{u})$ cancels out in the logarithm ratio, significantly simplifying the computation. The equation splits into two components:

(1) Expected Log-Likelihood:

$$\iint q(\mathbf{u}) p(\mathbf{f} | \mathbf{u}) \log p(\mathbf{y} | \mathbf{f}) d\mathbf{f}d\mathbf{u} = \mathbb{E}_{q(\mathbf{f})}[\log p(\mathbf{y} | \mathbf{f})] = \sum_{i=1}^N \mathbb{E}_{q(f_i)}[\log p(y_i | f_i)], \quad (\text{A.5})$$

where $q(\mathbf{f}) = \int p(\mathbf{f}|\mathbf{u})q(\mathbf{u})d\mathbf{u}$ represents the marginal variational distribution.

(2) KL Divergence:

$$\int q(\mathbf{u}) \left[\int p(\mathbf{f} | \mathbf{u}) d\mathbf{f} \right] \log \frac{p(\mathbf{u})}{q(\mathbf{u})} d\mathbf{u} = -\text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{A.6})$$

Note that $\int p(\mathbf{f}|\mathbf{u})d\mathbf{f} = 1$. Combining (A.5) and (A.6), we obtain the final ELBO formulation as presented in Equation (4):

$$\mathcal{L}_{\text{SGP}} = \sum_{i=1}^N \mathbb{E}_{q(f_i)} [\log p(y_i | f_i)] - \text{KL}[q(\mathbf{u}) \| p(\mathbf{u})]. \quad (\text{A.7})$$

B Proof of Unbiased Gradient Estimation via Reparameterization Trick

This appendix provides the mathematical justification for the gradient computation in the Doubly Stochastic Variational Inference (DSVI) framework, specifically concerning Equations (9)-(12) in Section 2.3. We aim to compute the gradient of the expected log-likelihood term in the ELBO with respect to the variational parameters ϕ (i.e., $\mathbf{m}, \mathbf{S}$). The objective function is:

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi(\mathbf{f})} [\log p(\mathbf{y} | \mathbf{f})] = \int q_\phi(\mathbf{f}) \log p(\mathbf{y} | \mathbf{f}) d\mathbf{f}. \quad (\text{B.1})$$

Directly differentiating (B.1) is problematic because the integration domain depends on ϕ . To resolve this, we apply the reparameterization trick defined in Equation (11):

$$\mathbf{f} = g(\phi, \epsilon) = \boldsymbol{\mu}_\phi + \boldsymbol{\Sigma}_\phi^{1/2} \odot \epsilon, \quad \epsilon \sim p(\epsilon) = \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{B.2})$$

Using the Law of the Unconscious Statistician (LOTUS), the expectation over $q_\phi(\mathbf{f})$ can be rewritten as an expectation over the auxiliary noise distribution $p(\epsilon)$, which is independent of ϕ :

$$\mathcal{L}(\phi) = \mathbb{E}_{p(\epsilon)} [\log p(\mathbf{y} | g(\phi, \epsilon))]. \quad (\text{B.3})$$

Now, the gradient operator ∇_ϕ can be moved inside the expectation:

$$\nabla_\phi \mathcal{L}(\phi) = \nabla_\phi \int p(\epsilon) \log p(\mathbf{y} | g(\phi, \epsilon)) d\epsilon = \mathbb{E}_{p(\epsilon)} [\nabla_\phi \log p(\mathbf{y} | g(\phi, \epsilon))]. \quad (\text{B.4})$$

This expectation can be efficiently approximated using Monte Carlo sampling. For S samples of $\epsilon_s \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$:

$$\nabla_\phi \mathcal{L}(\phi) \approx \frac{1}{S} \sum_{s=1}^S \nabla_\phi \log p(\mathbf{y} | g(\phi, \epsilon_s)). \quad (\text{B.5})$$

Since $g(\phi, \epsilon)$ is a deterministic differentiable function, the gradients can be computed via standard backpropagation (chain rule). This proves that the reparameterization trick provides an unbiased low-variance gradient estimator for optimizing the ELBO in Equation (9).

C Theoretical Justification of the Weighted ELBO

This appendix provides the theoretical basis for the sample-weighted ELBO objective function defined in Equation (16). In **standard** variational inference, the KL divergence between the variational posterior $q(\mathbf{u})$ and the true Bayesian posterior $p(\mathbf{u}|\mathbf{y}) \propto p(\mathbf{y}|\mathbf{f})p(\mathbf{f}|\mathbf{u})p(\mathbf{u})$ is minimized. This is equivalent to maximizing the standard ELBO:

$$\mathcal{L} = \mathbb{E}_q[\log p(\mathbf{y} | \mathbf{f})] - \text{KL}[q(\mathbf{u}) \parallel p(\mathbf{u})]. \quad (\text{C.1})$$

To incorporate the sample-level attention mechanism rigorously, we adopt the framework of generalized variational inference (Knoblauch et al., 2019). In GVI, the standard Bayesian inference is generalized as an optimization problem involving a loss function ℓ , a divergence D , and a prior π :

$$q^* = \arg \min_q \{ \mathbb{E}_q[\ell(\mathbf{y}, \mathbf{f})] + D(q \parallel \pi) \}. \quad (\text{C.2})$$

In our DA-DGP framework, we define a weighted log-likelihood loss to emphasize the ROI:

$$\ell_{\text{weighted}}(\mathbf{y}, \mathbf{f}) = - \sum_{k=1}^p \sum_{i=1}^N \tilde{s}_{i,k} \log p(y_{i,k} | f_{i,k}^L), \quad (\text{C.3})$$

where $\tilde{s}_{i,k}$ are the normalized attention weights derived in Equation (14). Setting the divergence D to be the KL divergence, the optimization objective becomes:

$$\mathcal{L}_{\text{DA-DGP}} = \sum_{k=1}^p \sum_{i=1}^N \tilde{s}_{i,k} \mathbb{E}_{q(f_{i,k}^L)} [\log p(y_{i,k} | f_{i,k}^L)] - \text{KL}[q(\mathbf{u}) \parallel p(\mathbf{u})]. \quad (\text{C.4})$$

This derivation confirms that our weighted objective (Equation 16) is not merely a heuristic modification but a valid GVI posterior approximation that minimizes a preference-weighted risk bound.

D Supplementary Visualizations of Experimental Results

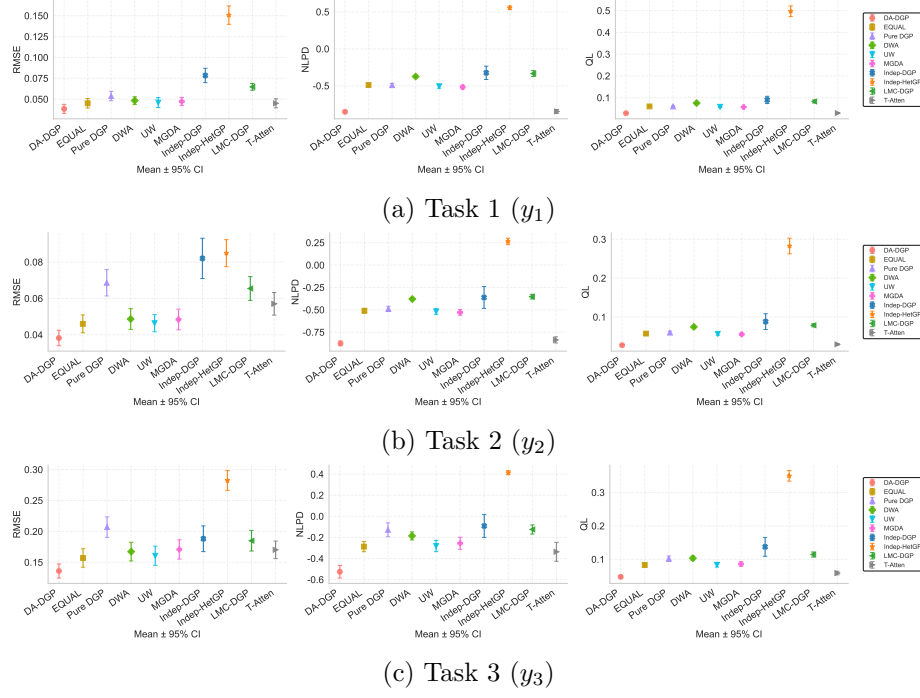


图 D.1: Auxiliary taskwise performance metric curves for the numerical example.

数值算例的辅助分任务性能指标曲线。

E Sensitivity Analyses for the Numerical Example

This appendix reports the detailed sensitivity analyses referenced in Section 4.1 of the main paper. We consider three factors: the Gaussian bandwidth used in sample-level attention, the surrogate depth/hidden complexity, and the available training sample size. All statistics are aggregated over 20 independent runs, and the figures below visualize taskwise error bars for RMSE, NLPD, and QL.

本附录给出正文 Section 4.1 引用的详细敏感性分析结果。我们考察三个因素：样本级注意力中的 Gaussian bandwidth、代理模型的深度/隐藏复杂度，以及可用训练样本量。所有统计量均基于 20 次独立运行汇总得到，下列图表进一步展示了各任务在 RMSE、NLPD 和 QL 上的误差棒结果。

E.1 Bandwidth Sensitivity

The bandwidth study confirms that moderate kernels are preferable. Averaged over three tasks, $\sigma = 0.5$ yields the best QL (**0.034774**), while $\sigma = 0.3$ attains the best RMSE (**0.069403**) and remains very close to $\sigma = 0.5$. In contrast, $\sigma = 0.1$ degrades both RMSE and QL, indicating over-localization of the training signal. Hence, the default choice $\sigma = 0.5$ provides the most balanced overall behavior.

带宽敏感性结果表明，中等带宽更为合适。在三个任务平均意义下， $\sigma = 0.5$ 取得最佳 QL (**0.034774**)，而 $\sigma = 0.3$ 取得最佳 RMSE (**0.069403**)，且与 $\sigma = 0.5$ 非常接近。相比之下， $\sigma = 0.1$ 会同时恶化 RMSE 和 QL，说明训练信号被过度收缩到 ROI 中极窄邻域。因此，默认设定 $\sigma = 0.5$ 在综合表现上最为均衡。

表 E.1: Average sensitivity summary for Gaussian bandwidth.

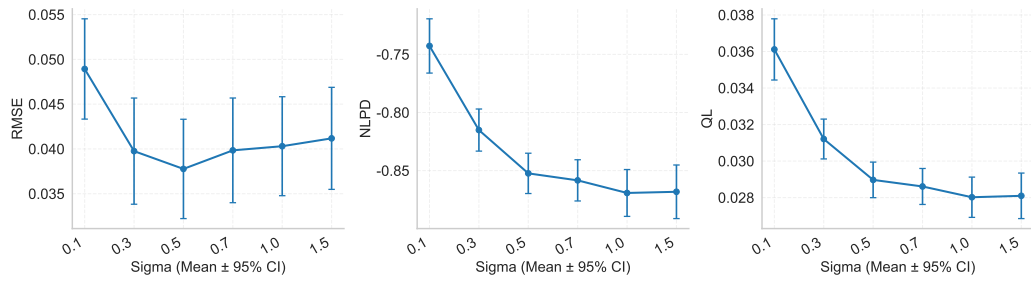
Gaussian 带宽的平均敏感性汇总。

Condition	Avg. RMSE	Avg. NLPD	Avg. QL
$\sigma = 0.1$	0.090653	-0.611964	0.044973
$\sigma = 0.3$	0.069403	-0.734488	0.035847
$\sigma = 0.5$	0.070831	-0.750523	0.034774
$\sigma = 0.7$	0.075690	-0.737188	0.035506
$\sigma = 1.0$	0.078762	-0.731904	0.035760
$\sigma = 1.5$	0.082363	-0.717340	0.036645

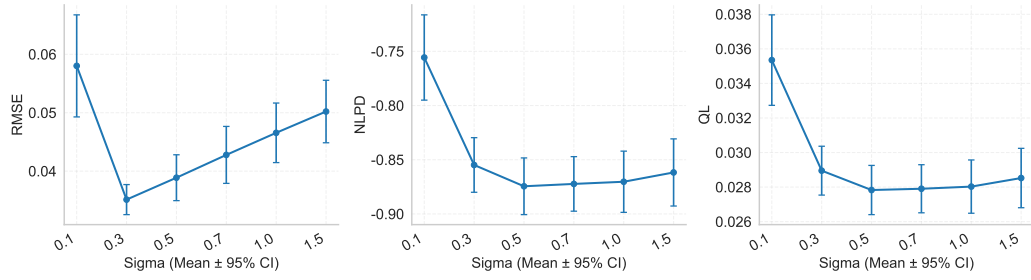
E.2 Depth and Hidden Complexity Sensitivity

The depth study shows that the default configuration is not only simpler but also clearly superior. The 2L-D5 model achieves the best average RMSE (**0.070808**), NLPD (**-0.751518**), and QL (**0.034732**). Shallower 2L-D2 underfits, while deeper 3L-D5, 4L-D5, and 5L-D5 rapidly deteriorate, suggesting that the current 5-dimensional benchmark with 400 training samples does not benefit from additional depth.

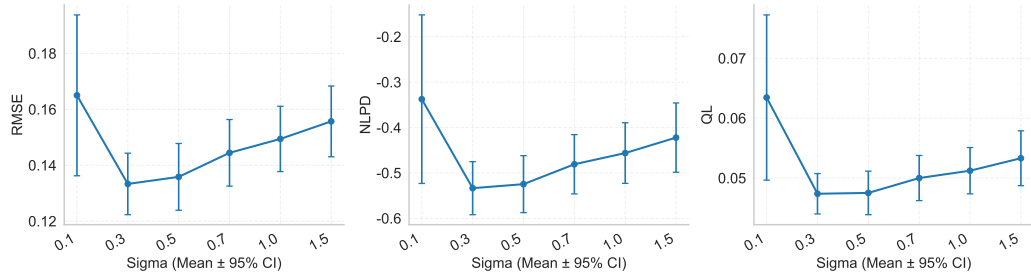
深度与隐藏复杂度敏感性表明，默认配置不仅更简单，而且性能也显著更优。2L-D5 在平均 RMSE (**0.070808**)、NLPD (**-0.751518**) 和 QL (**0.034732**) 上均为最佳。较浅的



(a) Task 1



(b) Task 2



(c) Task 3

图 E.1: Taskwise error-bar plots for Gaussian bandwidth sensitivity.

Gaussian 带宽敏感性的分任务误差棒图。

2L-D2 存在欠拟合，而更深的 3L-D5、4L-D5 和 5L-D5 则迅速恶化，说明当前 5 维、400 样本的基准问题并不能从额外深度中获益。

表 E.2: Average sensitivity summary for depth and hidden complexity.

深度与隐藏复杂度的平均敏感性汇总。

Condition	Avg. RMSE	Avg. NLPD	Avg. QL
2L-D2	0.246244	0.084328	0.185969
2L-D5	0.070808	-0.751518	0.034732
3L-D5	0.291672	0.405546	0.282074
4L-D5	0.460090	0.774311	0.650473
5L-D5	0.706363	1.235559	1.192501

E.3 Sample Size Sensitivity

The sample-size experiment highlights a strong finite-sample effect. Performance improves monotonically as the training size grows from 20 to 400. In particular, the average QL drops from 1.439658 at $n = 20$ to **0.034749** at $n = 400$, and the average RMSE drops from 0.592939 to **0.070777**. The pronounced gain from $n = 200$ to $n = 400$ further indicates that 400 samples are not excessive for this heterogeneous benchmark.

样本量敏感性实验揭示了明显的有限样本效应。随着训练样本量从 20 增加到 400，性能呈单调改善趋势。特别是，平均 QL 从 $n = 20$ 时的 1.439658 降至 $n = 400$ 时的 **0.034749**，平均 RMSE 也从 0.592939 降至 **0.070777**。其中从 $n = 200$ 到 $n = 400$ 的显著跃迁进一步说明，对于该异质基准问题而言，400 个样本并不冗余。

表 E.3: Average sensitivity summary for training sample size.

训练样本量的平均敏感性汇总。

Condition	Avg. RMSE	Avg. NLPD	Avg. QL
$n = 20$	0.592939	1.128905	1.439658
$n = 50$	0.444696	0.992585	1.129003
$n = 100$	0.403108	0.950088	1.033113
$n = 200$	0.200731	0.304830	0.285129
$n = 400$	0.070777	-0.751240	0.034749

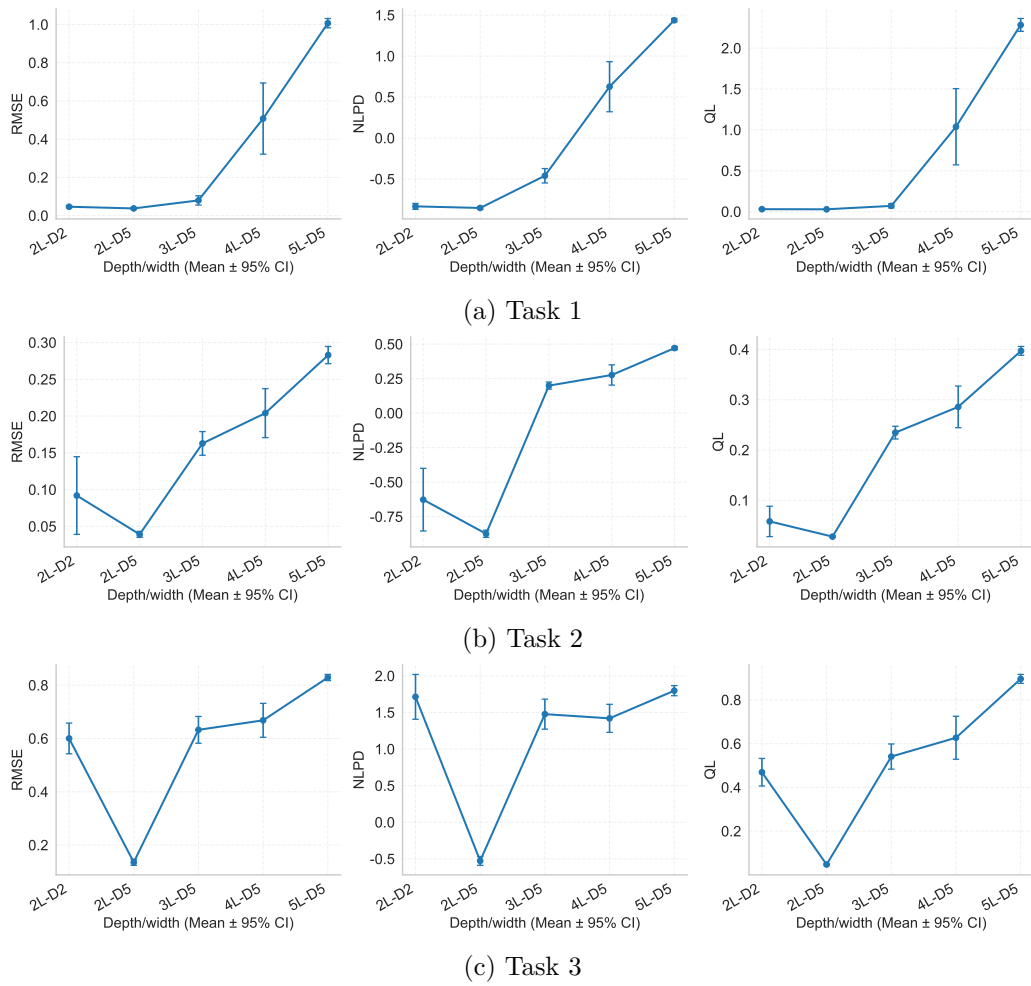


图 E.2: Taskwise error-bar plots for depth and hidden-complexity sensitivity.

深度与隐藏复杂度敏感性的分任务误差棒图。

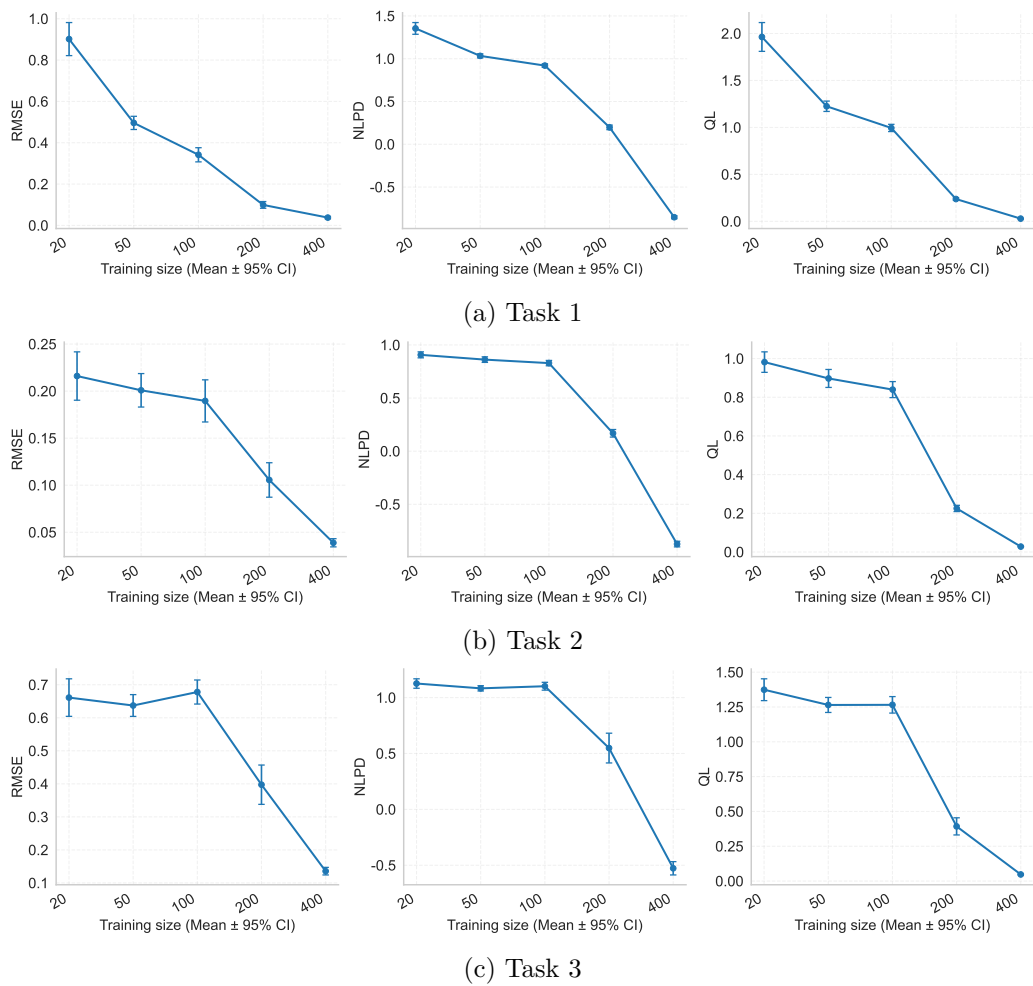


图 E.3: Taskwise error-bar plots for sample-size sensitivity.

训练样本量敏感性的分任务误差棒图。

F Wall-clock Training Time and Time Complexity

This appendix reports the wall-clock training time and asymptotic time-complexity order of the Gaussian process surrogate methods evaluated in Section 4.2 of the main paper. All methods are trained for 500 epochs on the OFDM dataset with $N = 400$ training samples, $B = 128$, hidden latent dimension $h = 5$, $M = 128$ inducing points, $p = 3$ response tasks, and $q = 3$ LMC latent functions where applicable, on a single workstation equipped with an NVIDIA RTX 2060 GPU, PyTorch 2.11+cu130 and Python 3.10. Table F.1 summarizes the total wall-clock time, per-epoch time, trainable-parameter count, and total asymptotic training order for each method.

本附录给出正文 Section 4.2 中各高斯过程代理方法的 wall-clock 训练耗时与渐近时间复杂度。所有方法在 OFDM 数据集上训练 500 epochs，使用 $N = 400$ 训练样本、 $B = 128$ 、隐藏潜在维度 $h = 5$ 、 $M = 128$ 个 inducing points、 $p = 3$ 个响应任务；对 LMC-DGP，latent functions 数为 $q = 3$ 。硬件为一块 NVIDIA RTX 2060 GPU，软件栈为 PyTorch 2.11+cu130、Python 3.10。Table F.1 汇总各方法的总 wall-clock 耗时、单 epoch 耗时、可训练参数量以及总训练渐近阶。

表 F.1: Wall-clock training time of all Gaussian process surrogate methods on the OFDM dataset (500 epochs, NVIDIA RTX 2060, PyTorch 2.11+cu130).

在 OFDM 数据集上各高斯过程代理方法的 wall-clock 训练耗时 (500 epochs, NVIDIA RTX 2060, PyTorch 2.11+cu130)。

Method	Total (s)	s/epoch	Params (K)	Total asymptotic order
DA-DGP	397.79	0.7956	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
T-Atten	387.78	0.7756	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
MGDA	164.87	0.3297	136.6	$\mathcal{O}(pE\lceil N/B \rceil(h+p)(BM^2+M^3))$
Indep-DGP	163.40	0.3268	306.9	$\mathcal{O}(pE\lceil N/B \rceil(h+1)(BM^2+M^3))$
DWA	109.25	0.2185	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
LMC-DGP	103.35	0.2067	136.7	$\mathcal{O}(E\lceil N/B \rceil(h+q)(BM^2+M^3))$
Equal	100.57	0.2011	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
Pure DGP	99.56	0.1991	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
UW	96.04	0.1921	136.6	$\mathcal{O}(E\lceil N/B \rceil(h+p)(BM^2+M^3))$
Indep-HetGP	10.51	0.0210	0.021	$\mathcal{O}(EpN^3)$

The table reports asymptotic orders up to constant factors, so DA-DGP and T-Atten have the same formal order as the shared DGP backbone. Their larger wall-clock times arise from constant-factor operations introduced by bilevel optimization: the inner virtual update, the unrolled validation backward, and two finite-difference Hessian–vector evaluations. The gradient-balancing baseline (MGDA) and the independent-task baseline (Indep-DGP) instead scale approximately linearly in the number of tasks p . The heteroscedastic exact GP (Indep-HetGP) bypasses the variational machinery and is dominated by the dense $N \times N$ Cholesky factorization, which remains affordable because $N = 400$ is small. These measurements quantify the extra wall-clock overhead of DA-DGP and show that it remains practical for a one-shot surrogate fit under the robust Pareto assessment reported in Table 7 of the main paper.

表中渐近阶省略常数因子，因此 DA-DGP 与 T-Atten 在形式上与共享 DGP 主干保持相同阶。二者 wall-clock 时间较高，来自 bilevel 优化引入的常数因子操作，包括内层虚拟更新、验证 unrolled 反向传播以及两次 finite-difference Hessian–vector 近似；梯度均衡基

线 (MGDA) 和独立任务基线 (Indep-DGP) 则近似随任务数 p 线性增长。异方差 exact GP (Indep-HetGP) 不经过变分机制, 其代价由 $N \times N$ Cholesky 分解主导, 因 $N = 400$ 较小而仍可接受。综上, 这些测量量化了 DA-DGP 额外的 wall-clock 开销, 并表明其在正文 Table 7 所示鲁棒 Pareto 评价下, 对于一次性代理拟合仍具有实际可行性。

F.1 Robust Pareto Indicator Settings

For reproducibility of the AHP-weighted TOPSIS ranking in Table 7 of the main paper, the nine robust Pareto indicators are treated as follows. The benefit-type indicators are the number of within-method Pareto points ($\#P$), the number of contributions to the global front ($\#GP$), global contribution rate (GC), normalized hypervolume (HV_n), and hypervolume ratio (HVR), whereas the cost-type indicators are inverted generational distance (IGD), inverted generational distance plus (IGD^+), spacing metric (SP), and closest-ideal distance (CID). The raw AHP weights assigned to these indicators are $[6, 3, 3, 1, 6, 1, 6, 1, 6]$, respectively, and are normalized to sum to one before TOPSIS aggregation. This setting gives higher emphasis to within-method frontier density, hypervolume ratio, IGD^+ , and CID while retaining the remaining indicators as supporting evidence, so the final C_i rank should be read as an integrated robust-frontier assessment rather than a claim of dominance on every single metric.

为便于复核正文 Table 7 的 AHP 加权 TOPSIS 排名, 九个鲁棒 Pareto 指标按如下方向处理: 方法内 Pareto 点数 ($\#P$)、对全局前沿的贡献数 ($\#GP$)、全局贡献率 (GC)、归一化超体积 (HV_n) 和超体积比 (HVR) 为收益型指标, 反世代距离 (IGD)、增强反世代距离 (IGD^+)、spacing metric (SP) 与最近理想点距离 (CID) 为成本型指标。对应的 AHP 原始权重依次为 $[6, 3, 3, 1, 6, 1, 6, 1, 6]$, 并在 TOPSIS 聚合前归一化为和为一。该设定对方法内前沿密度、超体积比、 IGD^+ 和 CID 赋予更高权重, 同时保留其余指标作为支撑证据, 因此最终 C_i 排名应理解为综合鲁棒前沿评价, 而非声称某一方法在每个单项指标上均逐项支配。

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